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Risk

- Measurable probability of outcomes
- Probabilities can be assigned to expected outcomes

Stock market investments

Uncertainty

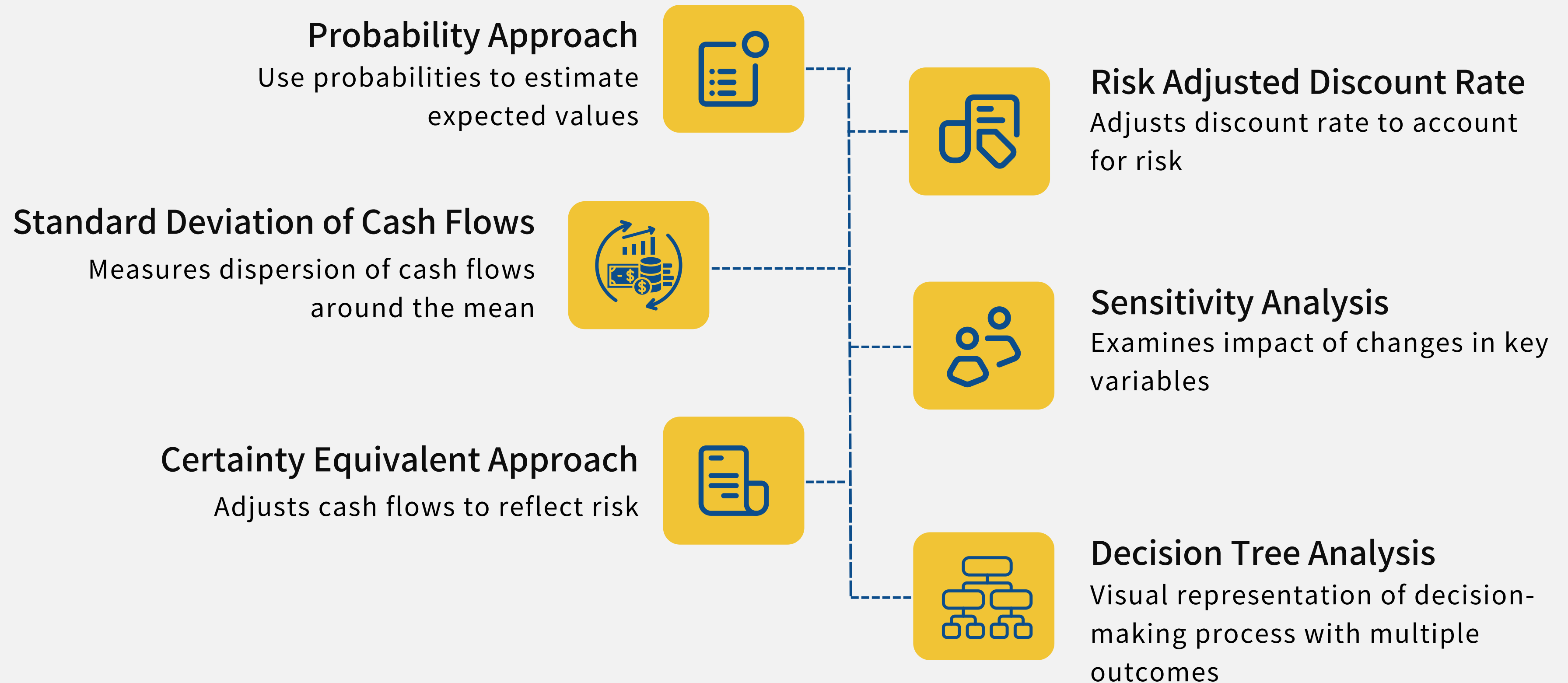
- Outcomes cannot be predicted with certainty
- Probabilities cannot be assigned

Investing in a new, untested technology

Risk & Uncertainty



Measures of Risk

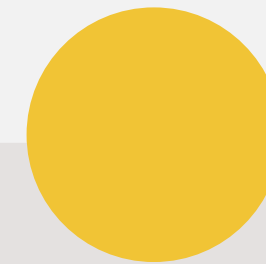


Probability Approach



Step 1

Calculate an expected value of the NPV



Step 2

Measure risk

- Calculate the worst possible outcome and its probability.
- Calculate the probability that the project will fail to achieve a positive NPV.
- Calculate the standard deviation of the NPV.

Question 1

ABC Ltd is evaluating a proposal having an initial outlay of Rs 140,000 and economic life of 2 years. The cash inflows and the respective probabilities have been found to be as follows.

Evaluate the proposal given that the firm has minimum required rate of return of 10%.

Year 1		Year 2	
Cash Inflow	Probability	Cash Inflow	Probability
100,000	0.3	40,000	0.5
80,000	0.5	70,000	0.3
10,000	0.2	60,000	0.2

Solution

Estimated cash flow for Year 1:

$$CI \text{ Yr 1} = 100,000 \times 0.3 + 80,000 \times 0.5 + 10,000 \times 0.2$$

$$CI \text{ Yr 1} = 30,000 + 40,000 + 2,000$$

$$CI \text{ Yr 1} = 72,000$$

Estimated cash flow for Year 2:

$$CI \text{ Yr 2} = 40,000 \times 0.5 + 70,000 \times 0.3 + 60,000 \times 0.2$$

$$CI \text{ Yr 2} = 20,000 + 21,000 + 12,000$$

$$CI \text{ Yr 2} = 53,000$$

	T0	T1	T2
Cash Flow	-140,000	72,000	53,000
DF @ 10%	1.000	0.909	0.826
PV	-140,000	65,448	43,778
NPV	-30,774		

Question 3

XYZ Ltd is evaluating two equal size mutually exclusive proposals A and B for which the respective cash flows with associated probabilities are as follows.

Find out the risks of the proposal in terms of standard deviation and evaluate the proposals.

A		B	
Cash flows	Probability	Cash flows	Probability
2000	0.3	1000	0.1
4000	0.4	3000	0.1
6000	0.3	5000	0.4
		7000	0.3
		9000	0.1

Solution

Year 1:

Cash flow, x	Probability, p	x * p	x^2 * p
2000	0.3	600	12,00,000
4000	0.4	1600	64,00,000
6000	0.3	1800	1,08,00,000
Total	1	4000	1,84,00,000

Expected value = 4,000

$$\text{Standard deviation} = \sqrt{1,84,00,000 - (4000)^2}$$

Standard deviation = 1549.19

Coefficient of Variation (CV):

$$CV = \frac{\text{Standard deviation}}{\text{Expected value}} \times 100\%$$

$$CV = (1549.19 / 4000) \times 100\%$$

CV = 38.73%

Year 2:

Cash flow, x	Probability, p	x * p	x^2 * p
1000	0.1	100	1,00,000
3000	0.1	300	9,00,000
5000	0.4	2000	1,00,00,000
7000	0.3	2100	1,47,00,000
9000	0.1	900	81,00,000
Total	1	5400	3,38,00,000

Expected value = 5,400

Standard deviation = $\sqrt{(3,38,00,000 - (5400)^2)}$

Standard deviation = 2154.07

Coefficient of Variation (CV):

CV = (SD / EV) x 100%

CV = (2154.07 / 5400) x 100%

CV = 39.89%

Results:

	Proposal A	Proposal B
Expected value	4000	5400
Standard deviation	1549.19	2154.07
Coefficient of variation	38.73%	39.89%

Certainty-Equivalent Approach

- Expected cash flows of the project are converted to riskless equivalent amounts
- Greater the risk of an expected cash flow,
 - the smaller the certainty-equivalent value (for revenue / cash inflow)
 - the larger the certainty equivalent value (for expenses / cash outflow)
- Cash flow discounted at a **risk free rate**

Practice Question

Golden Era Inc., whose cost of capital is 12%, is considering a project with the following expected cash flows.

The project seems to be worthwhile. However, because of the uncertainty about the future cash receipts, the management decides to reduce them to 'certainty-equivalents' by taking only 70%, 65%, and 60% of the years 1, 2, and 3 cash flows respectively. The risk-free rate is 6%.

On the basis of the information set out above, assess whether the project is worthwhile.

Year	Cash Flow (\$)	Discount Factor @ 12%	Present Value (\$)
0	(15,000)	1.000	(15,000)
1	10,000	0.893	8,930
2	8,000	0.797	6,376
3	6,000	0.712	4,272
		NPV =	4,578

Year	Cash Flow (\$)	Discount Factor @ 6%	Present Value (\$)
0	(15,000)	1.000	(15,000)
1	$10,000 \times 0.7 = 7,000$	0.943	6,601
2	$8,000 \times 0.65 = 5,200$	0.890	4,628
3	$6,000 \times 0.6 = 3,600$	0.840	3,024
		NPV =	-747

Question 7

ABC and co is considering 2 mutually exclusive machines x and y. the estimated cashflows and certainty equivalents for both the machine are as follows :

Machine X

Machine Y

Year	Cash Flow	Certainty Equivalent	Cash Flow	Certainty Equivalent
0	-30000	1	-40000	1
1	15000	0.95	25000	0.9
2	15000	0.85	20000	0.8
3	10000	0.7	15000	0.7
4	10000	0.65	10000	0.6

Machine X

Year	Cash Flow	DF @ 5%	Present Value (PV)
0	$-30,000 \times 1 = -30,000$	1	$-30,000 \times 1 = -30,000$
1	$15,000 \times 0.95 = 14,250$	0.952	$14,250 \times 0.952 = 13,566$
2	$15,000 \times 0.85 = 12,750$	0.907	$12,750 \times 0.907 = 11,564$
3	$10,000 \times 0.70 = 7,000$	0.864	$7,000 \times 0.864 = 6,048$
4	$10,000 \times 0.65 = 6,500$	0.823	$6,500 \times 0.823 = 5,350$
		NPV =	6,528

Machine Y

Year	Cash Flow	DF @ 5%	Present Value (PV)
0	$-40,000 \times 1 = -40,000$	1	$-40,000 \times 1 = -40,000$
1	$25,000 \times 0.9 = 22,500$	0.952	$22,500 \times 0.952 = 21,420$
2	$20,000 \times 0.8 = 16,000$	0.907	$16,000 \times 0.907 = 14,512$
3	$15,000 \times 0.7 = 10,500$	0.864	$10,500 \times 0.864 = 9,072$
4	$10,000 \times 0.6 = 6,000$	0.823	$6,000 \times 0.823 = 4,938$
		NPV =	9,942

Risk Adjusted Discount Rate (RADR)

$$RADR = R_f + Risk\ Index \times (R_c - R_f)$$

R_f = Risk free rate

R_c = Company's Cost of Capital

Practice Question

TechPioneers Ltd. is considering a new project with an initial investment of \$200,000. The project has a beta of 1.1, the risk-free rate is 4%, and the expected market return is 11%. Calculate the **risk-adjusted discount rate**.

$$RADR = R_f + Risk\ Index \times (R_c - R_f)$$

$$RADR = 4\% + 1.1 \times (11\% - 4\%)$$

$$RADR = 4\% + 1.1 \times 7\%$$

$$RADR = 4\% + 7.7\%$$

$$\mathbf{RADR = 11.7\%}$$

Practice Question

Future Innovations Corp. is evaluating a new project and has provided the following financial details. The project has a beta of 0.9, a risk-free rate of 3%, and the expected market return is 9%.

The anticipated cash flows for the project are as follows: an initial investment of \$150,000, followed by cash inflows of \$60,000 in Year 1, \$70,000 in Year 2, and \$80,000 in Year 3.

Calculate the **risk-adjusted discount rate** and determine the **Net Present Value (NPV)** of the project. Based on the NPV, assess whether the project is worthwhile.

$$RADR = R_f + Risk\ Index \times (R_c - R_f)$$

$$\text{Risk-Adjusted Discount Rate} = 3\% + 0.9 \times (9\% - 3\%)$$

$$RADR = 3\% + 0.9 \times 6\%$$

$$RADR = 3\% + 5.4\%$$

$$\mathbf{RADR = 8.4\%}$$

Year	Cash Flow (\$)	Discount Factor @ 8.4%	Present Value (\$)
0	-150,000	1.000	-150,000
1	60,000	0.923	55,380
2	70,000	0.851	59,570
3	80,000	0.785	62,800
		NPV =	27,750

Sensitivity Analysis

- Assesses how responsive the project's NPV is to changes in the variables used to calculate that NPV
- NPV depend on various independent variables:
 - Initial cost
 - Revenue or Sales (Selling price and Sales volume)
 - Operating costs
 - Cost of capital (Discount rate)

Sensitivity Analysis

$$\text{Sensitivity} = \frac{NPV}{PV \text{ of project variable}} \%$$

Lower the percentage, the more sensitive is NPV to that project variable

Variable would need to change by a smaller amount to make the project non-profitable

Question 11

Initial cash outflows: Rs. 1,00,000

Life: 4 years

Cash inflow per annum: Rs.40,000

Discount rate: 10%

Compute NPV and measure the sensitivity of the project to:

- A) Cost of the project
- B) Cash inflows
- C) Life
- D) Discount factor

Year	Cash Flow	Discount Factor @ 10%	Present Value
0	-1,00,000	1.000	-1,00,000
1	40,000	0.909	36,360
2	40,000	0.826	33,040
3	40,000	0.751	30,040
4	40,000	0.683	27,320
		NPV =	26,760

Sensitivity of the project to Cost of the project:

$$\text{Sensitivity} = 26,760 / 1,00,000 = \mathbf{26.76\%}$$

Sensitivity of the project to Cash inflows:

$$\text{Sum of PVs of cash flow} = 36,360 + 33,040 + 30,040 + 27,320 = 1,26,760$$

$$\text{Sensitivity} = 26,760 / 1,26,760 = \mathbf{21.11\%}$$

Sensitivity of the project to Life:

$$\text{Discounted payback period} = 3 + 560/27,320 = 3.02$$

$$\text{Sensitivity} = (3.02 / 4) - 1 = \mathbf{-24.49\%}$$

Sensitivity of the project to Cost of Capital:

$$\text{IRR} = 21.86\%$$

$$\text{Sensitivity} = (21.86\% / 10\%) - 1 = \mathbf{118.62\%}$$

PV	Cumulative PV
-1,00,000	-1,00,000
36,360	-63,640
33,040	-30,600
30,040	-560
27,320	26,760

